



SHAHJALAL UNIVERSITY OF SCIENCE AND TECHNOLOGY

SWE-420

Internship Report

Submitted By
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IICT, SUST

Performed At



Luminous Labs

Road No 11, Ave 6,
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LETTER OF TRANSMITTAL

July 12, 2022

The Director,

Institute of Information & Communication Technology,

Shahjalal University of Science & Technology.

Dear Sir,

I would like to submit my Internship report. This report is based on the internship program I participated from September 2021 to February 2022 in Luminous Labs, Bangladesh. This report is to show my learning outcome in the past 6 months. During this time, I worked under the supervision of Moniruz Zaman, CEO of Luminous Labs. It is my pleasure to present to you my work & experience with the industrial culture of Bangladesh.

Sincerely yours,

Taosif Sadly,

Reg No: 2017831026,

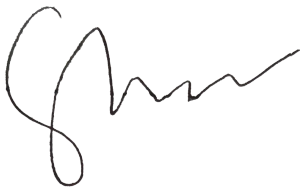
Dept. Of Software Engineering,

Shahjalal University of Science & Technology.

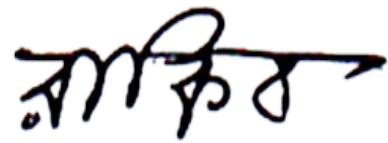
LETTER OF ENDORSEMENT

Subject: Approval of the report

This letter is to certify that content in this report is true & does not violate company confidentiality. The works mentioned in this report are authentic & have successful involvement of Taosif Sadly. I wish him a successful career.



Moniruz Zaman,
Chief Executive Officer,
Luminous Labs



Rakib Hossain,
Senior Software Engineer,
Luminous Labs

ACKNOWLEDGEMENT

It is my pleasure to present this report on my 6 months long internship period at Luminous Labs. I would like to express my gratitude to the Institute of Information & Communication Technology, SUST for arranging this opportunity, so I could learn first hand about using my knowledge on industry level works. I would also like to thank Luminous Labs for giving me the opportunity to work with them.

I would like to thank my honourable teacher, Fazle Mohammed Tausif, Assistant Professor, SWE-SUST, whose tireless effort got me this opportunity.

From Luminous Labs, I would like to thank Moniruz Zaman, CEO of Luminous Labs, who supported me, motivated me, guided me throughout this program.

I would also like to thank my Team Leader, Rakib Hossain, who taught me, helped me, mentored me & guided me in this period.

Finally, I would like to thank all members of Luminous Labs who are not mentioned above, whose kind welcome, help & teaching kept me going throughout this program.

EXECUTIVE SUMMARY

The internship report is to summarise the most significant achievements of the author at this 6month long program. The scope of this program is to identify the experience gained, projects done by the author in this program as an intern.

Institute of Information & Communication Technology, Shahjalal University of Science & Technology offers a full semester internship at 7th semester opportunity to its students. This program is to make sure the students get real life experience of the software industry before they get their certificate as a Software Engineer.

The author completed his internship at Luminous Labs. This program was from September 2021 to February 2022. The author got introduced to some new technology, programming language, and new frameworks throughout this program. But the most significant achievement for the author was the experience. He was introduced with professionalism, the software industry of Bangladesh.

The projects done by the author of this report increased his practical knowledge by a lot. He got some hands-on experience with some web & mobile application development technologies. Besides coding experience, completion of the tasks assigned to him taught the author about time efficiency, professionalism & made him confident about his ability.

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Chapter 1: Introduction

1.1 Introduction

According to [wikipedia](https://en.wikipedia.org/wiki/Internship), “An internship is a period of work experience offered by an organisation for a limited period of time”. Different institutions in Bangladesh offer internships to their students. Institute of Information & Communication Technology, SUST offers its students an internship program at their 7th semester in different recognised software farms of Bangladesh too. This program is to give the future software engineer a glance of the industry they are going to rock in the future. It’s a chance for the students to get introduced to their work environment before graduation. A glance at the exciting work of finally putting our knowledge into something used by real people. Institute of Information & Communication Technology, SUST always wants their students industry ready before graduation. This program is one of the last steps of achieving that.

I was sent to Luminous Labs, a Spain based Software Company that specialises on drop-shipping softwares, to complete my internship program. And it is my pleasure to say that I did the best I can to accomplish the purposes of this program. In this report, I will try to present to you the most important parts of my journey.

1.2 Objectives

This report is prepared as per the requirement provided to us by Institute of Information & Communication Technology, SUST as a part of the Internship program. This report is focused on my professional growth, learning outcome, project involvement throughout this semester.

1.3 Limitations

This report cannot show all of my works in the program as publishing some of them violates company privacy policy. It only contains the summary of all of my experience at Luminous Labs & some reviews from my mentors.

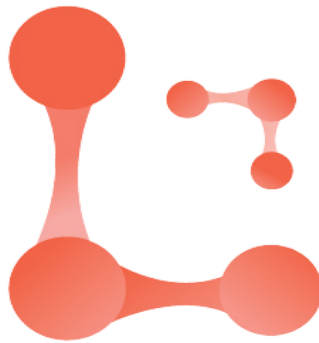
Chapter 2: Company Profile

In this section, I'll give an overview of the software farm, Luminous Labs, that I completed my internship in. This section contains the work range of Luminous Labs, its culture, activities, workflow etc.

2.1 About Luminous Labs

Luminous Labs is a Software Development Company with a vision to provide enterprise solutions to local and international organisations.

Luminous Labs started its journey in Oct. 2016. It is an independent software development and consulting company. As a professional software development company, Luminous Labs is highly concerned with the quality of its solutions and services.



The company was founded by a team of enthusiastic IT specialists who wanted to overcome the routine and create a company that would act in the market not only for business success but for the sake of IT itself. Thus, the mission of the company was defined to contribute to forward looking transformation of the society through software development.

Luminous Labs provides Software Security & Software Development for Business Automation, Business strategy development by adopting new Technology which includes Web Based Application Development, Software & Android Apps Development and Internet Marketing Strategy Development along with any kind of Web Application design.

Luminous Labs expertises in Drop-Shipping web applications, Mobile Application development, Education ERP Software, Hospital Management system etc.

2.2 Management

Luminous Labs management body has Moniruz Zaman as CEO, Ponkoj Shuvo as Manager.



Moniruz Zaman
CEO



Ponkoj Shuvo
Manager

2.3 Workflow

Luminous Labs has a good strategy to maintain a stable & nice working environment & to deliver products of the best quality. We have separate teams for every step of development. First, our analysis team analyses the project, understands the requirements of the project, makes time estimations etc. Then the project is handed over to a development team who will make the application. A team leader is assigned to each team to guide junior & other developers so the development process is smooth. While the development process is going on, the QA team tests the project continuously to ensure the best quality. The interesting fact in this process is, though the project is handed to the developers after proper understanding & planning, developers are free to consult with the customer any time of the development process. This makes things easier for the developer as he can question the client, make suggestions if he finds anything irrelevant or unnecessary or needs to change, make the client see the way he finds relevant. It's also helpful for the client, as he can always get good suggestions.

2.4 Clients

Luminous Labs has clients in 5 countries in the world. Some major Internet Service Providers in Bangladesh use our ISP solutions. Luminous Labs has worked with Ecommerce software, Education ERP Software, HR & Payroll management Software, Food delivery software etc. Some of their happy clients are Dropmatix, Droptienda, Dealox, Orbit, Buyer Seller etc.



2.5 Office Culture

Luminous Labs has a nice working environment. Employees are supposed to work 8 hours a day. Luminous Labs is strict about these hours. Working hour starts strictly at 10am & ends at 6pm. But Luminous Labs provides employees enough time for prayer, lunch, snacks etc. Luminous Labs is very flexible about leaves though. Employees are allowed to take leave on any kind of emergency without much hassle. We also have good work from home facilities (for a short period of time) in case an employee can't come to office physically for a time being. Luminous Labs provides employees lunch & snacks at the office.

2.6 Tools & Technology

Luminous labs is mainly a PHP based company. It expertises in the PHP framework, Laravel. But we are not necessarily stuck in a particular technology. Tools & Technologies used by Luminous are described below.

Languages

1. PHP
2. Java
3. Swift
4. Dart
5. GoLang
6. Javascript
7. Typescript
8. Python

Frameworks

1. Laravel
2. ReactJS
3. VueJS
4. NodeJS
5. Flutter

Database

1. SQL
2. MySQL
3. MongoDB



2.7 Human Resources

Luminous Labs has 25 employees. And their recruitment process is ongoing now. In the recruitment process, Luminous Labs emphasises problem solving.

Luminous Labs has an exciting culture of knowledge sharing. We get together weekly & share what we learn throughout this week.

2.8 Recreation

Luminous Labs regularly arranges recreation events like picnic, tour, sports events etc. to keep morale up among employees. We have indoor games like chess, carrom inside the office too so developers can enjoy their free times. There is a mini gym inside the office.





Chapter 3: Internship Activities

3.1 Activity Timeline

2021

JOINED LUMINOUS LABS

2021

STARTED MY FIRST
PROJECT WITH FLUTTER

2021

STARTED WORKING ON
LUMINOUS ISP

2021

STARTED DEVELOPING
MAP MODULE

2021

FINISHED MAP MODULE

2021

STARTED WORKING ON
DROPMATIX

2022

FINISHED ORBIT

2021

STARTED LEARNING
FLUTTER

2021

FINISHED LUMINOUS
E-BUY

2021

FINISHED LUMINOUS ISP

2021

STARTED LEARNING
GOLANG

2021

STARTED LEARNING
LARAVEL

2022

GOT ASSIGNED AS TEAM
LEADER FOR PROJECT...

2022

COMPLETED INTERNSHIP
AT LUMINOUS LABS

4 of us joined Luminous Labs as Interns. We were all assigned to separate teams to work on different projects. I was assigned to learn & work on cross platform mobile application development. I was assigned under the supervision of company CEO, Muniruz Zaman, who guided me in this stage. I was instructed to learn a Dart based framework, Flutter. After a brief learning period of 3weeks, my actual coding phase started.

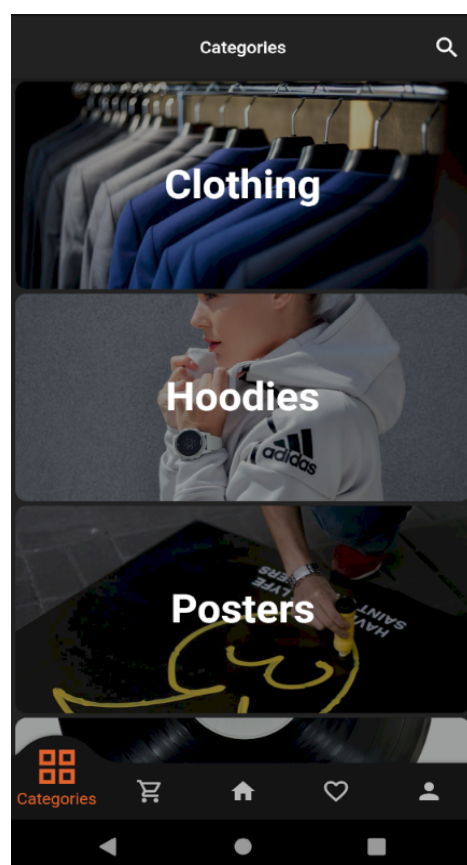
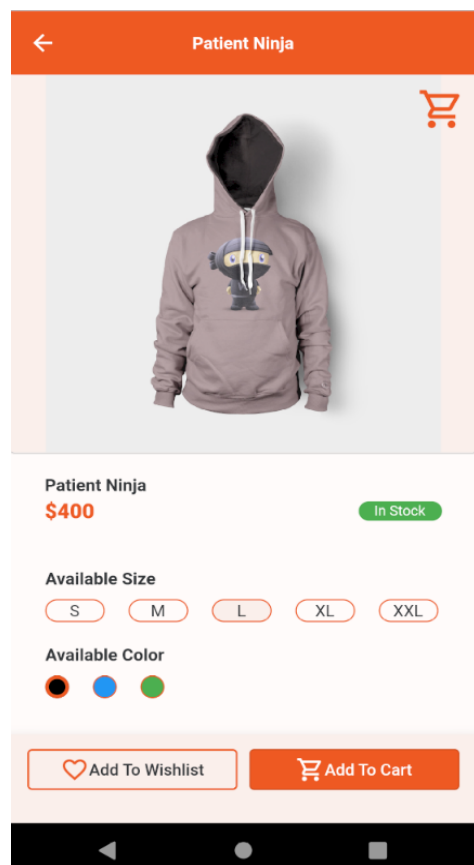
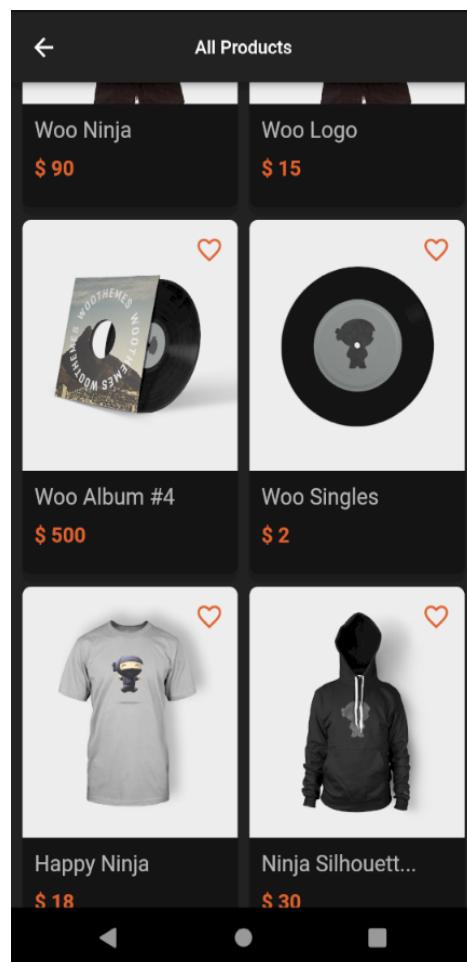
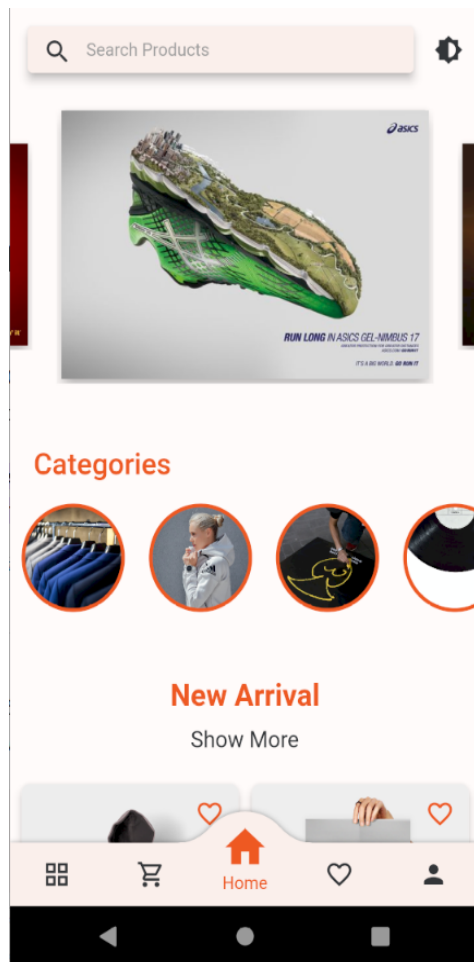
3.2 Project Participations

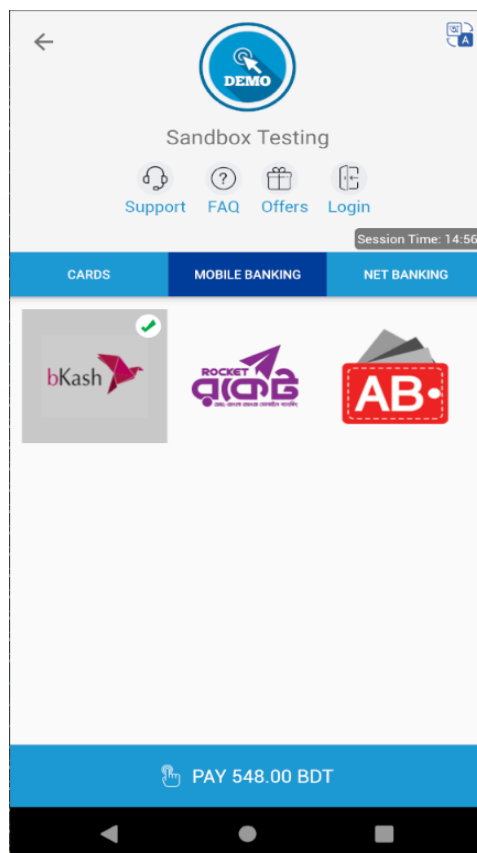
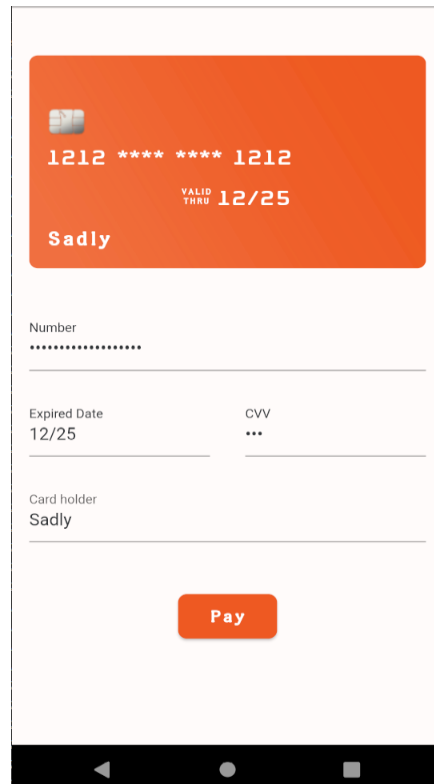
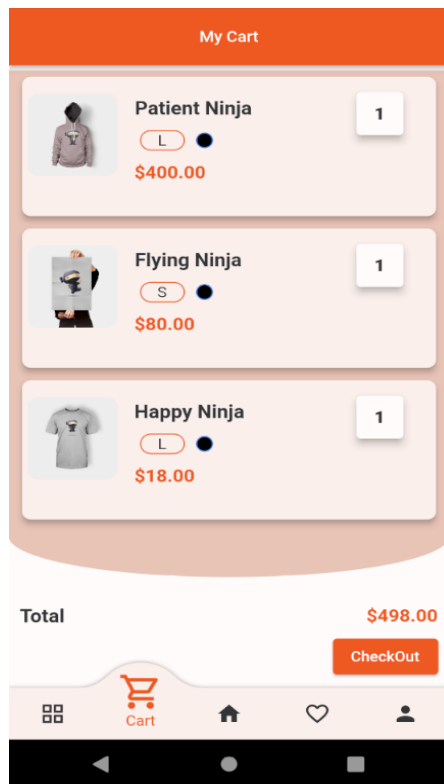
Luminous E-Buy

Luminous E-Buy is an ecommerce template for android I was assigned to develop. It is a multi vendor ecommerce mobile application built with flutter, laravel & mysql, which is intended to be used as a ready template to make applications for actual E-Commerce vendors. This app contains all functionality of an e-commerce app like Social Authentications, product browsing, searching, shop facilities, placing orders, order tracking etc. This app also has payment gateways like SSLCommerze, Stripe integrated. This template will be published for sale shortly in CodeCanyon. I worked alone on this project under the supervision of Moniruz Zaman.

Technologies used for this project are:

1. Flutter
2. Firebase
3. Laravel
4. Woocommerce Rest API v2
5. Google Map
6. MySQL
7. Different Packages





Google Map Module

This is a mobile application module intended for my learning & future usage. In this project I learned more about google map integration, customising google map, usage of polylines on google map, socket integration etc. This module is capable of tracking one or multiple users, connecting them through sockets, giving them direction towards one another, setting polylines to restrict an area on the map etc.

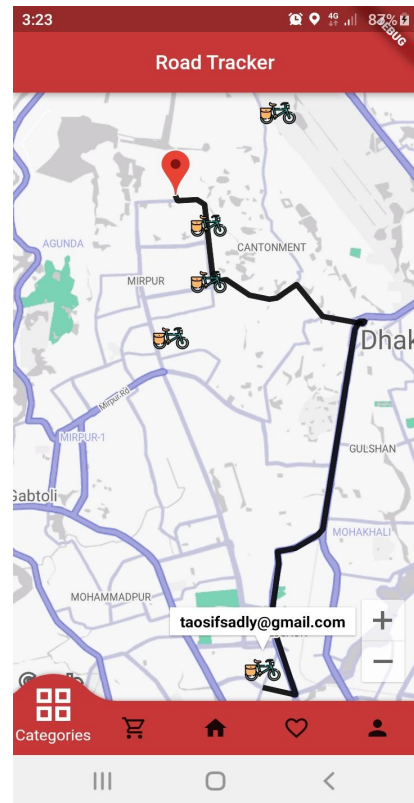
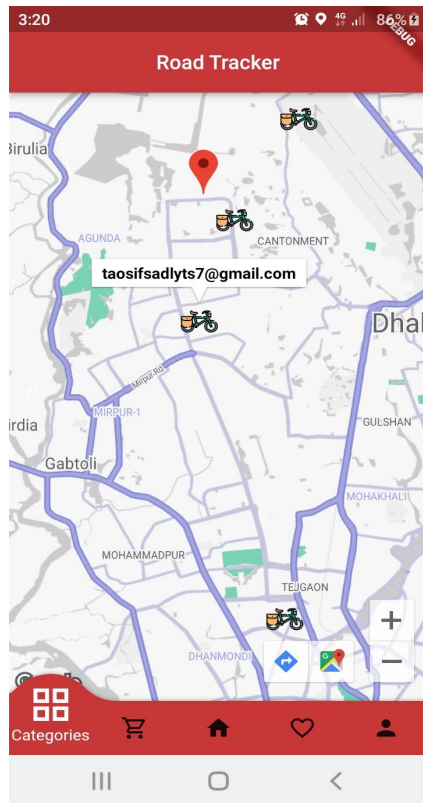
Here, I used flutter for the app development, integrated free & paid APIs from google map.

To implement connection between users, socket connection was implemented between them. A socket is one endpoint of a two-way communication link between two programs running on the network. A socket is bound to a port number so that the TCP layer can identify the application that data is destined to be sent to.

To implement socket connection between users, I used Socket.io. Socket.IO is a library that enables low-latency, bidirectional and event-based communication between a client and a server. I was instructed to use both GoLang & NodeJS & make 2 versions of Socket.io integration for this module. This functionality can be used in many applications, for example, food delivery. The user can be connected to his delivery man & track his current location at any time.

The module can also give direction for the user to any location he wants to go to. Users can search a place, select a destination & the app will show the shortest way to reach that destination.

This module is able to restrict an area on the map by setting polylines through some point & by getting the user's current location, decide whether the user is in an area or not. This functionality is useful for many purposes. For example, a system may want to send notifications to a user based on his location, like if he is inside a certain area, he will get a certain notification, but if he is not, he will not get any notification at all.

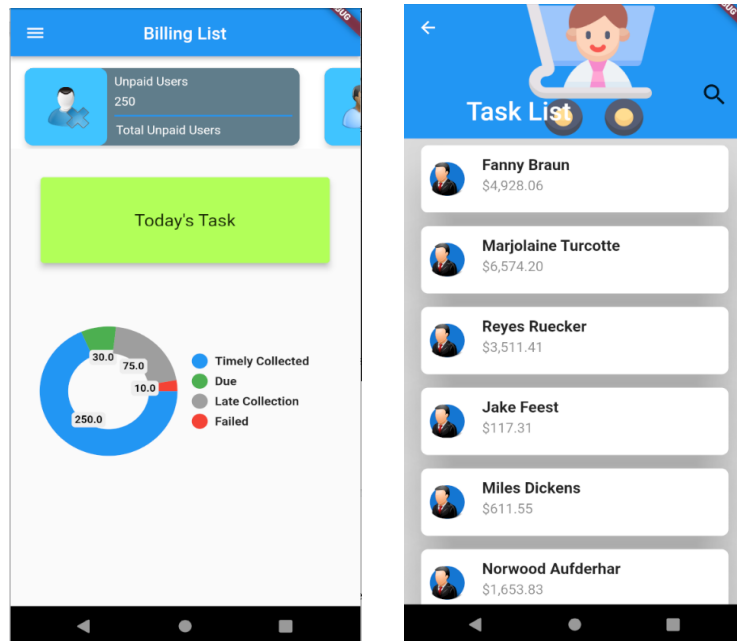


Technologies used in this project are:

1. Flutter
2. GoLang
3. NodeJS
4. Socket.io
5. Google Map APIs

Luminous ISP

It is a mobile application built with flutter, laravel & mysql that will automate bill collection activities for Internet Service Providers. This app is intended to be bundled with a web application for Internet Service Provider Luminous Labs is currently working on.



Technologies used for this project are:

1. Flutter
2. Laravel
3. MySQL

DropMatix



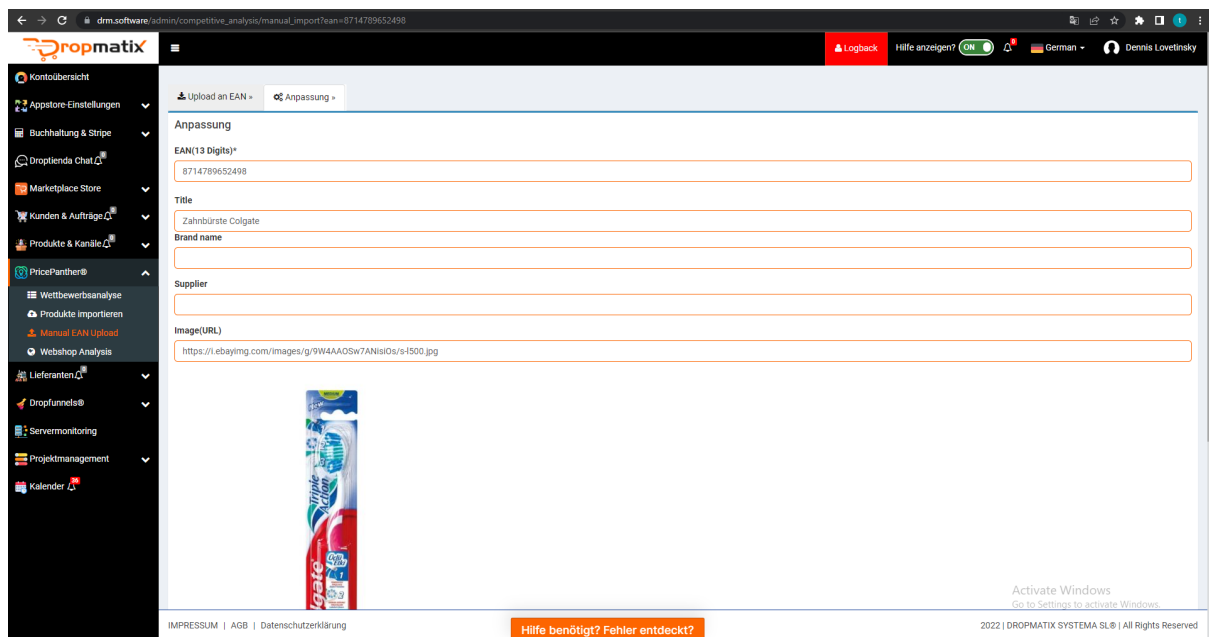
After some time working on mobile applications, I was assigned to DropMatix as a full-stack developer. There I was in a 5 member team that was led by Rakib Hossain, Senior Software Engineer.

DropMatix is a German based drop-shipping web application that deals with all kinds of drop shipping procedures. Drop-shipping is where a seller sets up a website and sells products that the seller does not keep in stock. When an order is placed, the seller sends it to a third party, the manufacturer, another retailer, or a wholesaler, who ships the goods directly to the buyer.

The part I worked on DropMatix, Competitive Analysis, was mostly analytical. Functionality of this part was, users can select some product from his shop for a global analysis. After selecting a product for Competitive Analysis, users will get updates weekly & monthly on different prices of that product on other shops on DropMatix. He can also see prices on other websites like EBay, Amazon, Google Shopping etc. These were the basic requirements of the part I worked on DropMatix, with many more additional functionalities. For these, I had to use EBay, Amazon, Google's APIs from other third party services.

The screenshot displays the DropMatix web application interface. The top navigation bar includes a 'Logback' button and a 'Hilfe anzeigen?' (Show help?) link. The main content area features a table with columns for product details and competitive pricing. The table is filtered by 'Gruppen/Intervalle' and 'Kanalfarben'. The table header includes columns for 'Titel/EAN', 'Marke', 'Lieferant', 'Quelle', 'Warengruppe', 'Bild', 'EK', 'VK-Preis', and 'UVP'. The table body lists several products, including 'Longboard Detroit Sur', 'Fahrradständer Shima', 'Elektrisches Fahrrad V', 'Kühl-Rucksack 1096', 'Taschenlampe Petzl T1', and 'Schutz der Gelenke bei Stürzen Tony Hawk Kid'. The table also includes a 'googleshopping' column with 'Bewertungen' (Reviews) and 'Preis' (Price). The bottom of the page shows a footer with 'IMPRESSUM | AGB | Datenschutzerklärung' and a 'Hilfe benötigt? Fehler entdeckt?' (Need help? Found a bug?) link.

Titel/EAN	Marke	Lieferant	Quelle	Warengruppe	Bild	EK	VK-Preis	UVP	googleshopping	Bewertungen	Preis
Longboard Detroit Sur EAN: 5056245573934	Sphero	dropmatix	MarketPlace Transfer	One Time		77,05€	0,00€	92,83€	124,01€	★★★★★	102,38€
Fahrradständer Shima EAN: 0022255968247	Livcell	dropmatix	MarketPlace Transfer	One Time		127,23€	0,00€	153,20€	167,33€	★★★★★	167,33€
Elektrisches Fahrrad V EAN: 8434127013391	IK	dropmatix	MarketPlace Transfer	One Time		1.618,17€	0,00€	1.703,34€	2.502,91€	★★★★★	2.502,91€
Kühl-Rucksack 1096 EAN: 3795464810018	Bimeda	dropmatix	MarketPlace Transfer	One Time		20,00€	0,00€	25,00€	25,00€	★★★★★	25,00€
Taschenlampe Petzl T1 EAN: 3342540827799	Victorinox	dropmatix	MarketPlace Transfer	One Time		18,49€	0,00€	22,83€	21,95€	★★★★★	23,63€
Schutz der Gelenke bei Stürzen Tony Hawk Kid EAN: 5056245597473	Vivacy	dropmatix	MarketPlace Transfer	One Time		24,23€	0,00€	29,55€	42,90€	★★★★★	39,74€



This part of my internship was completely on a PHP based framework, Laravel. Technologies used on this project are:

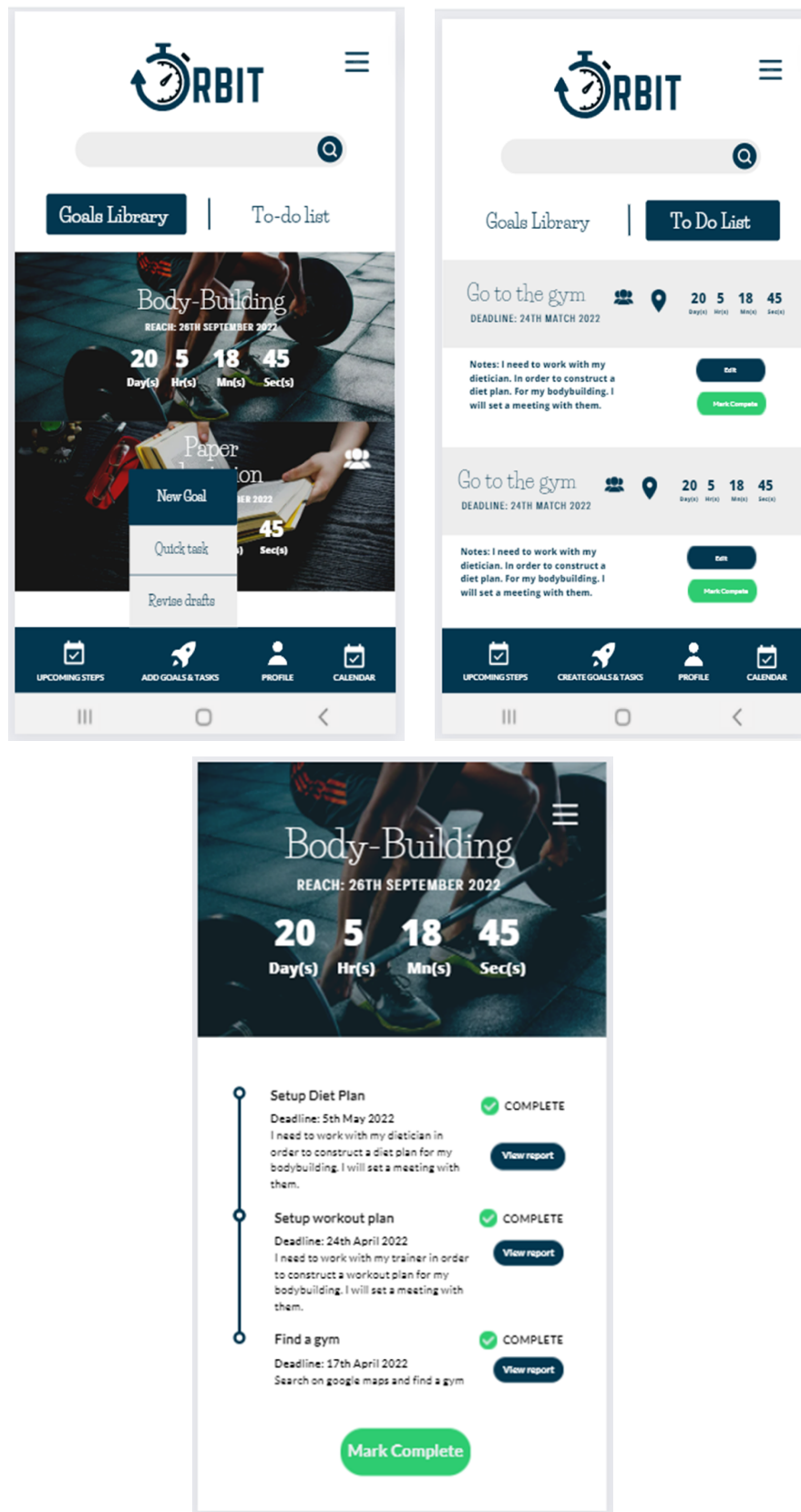
1. Laravel
2. Blade
3. HTML
4. CSS
5. JQuery
6. Bootstrap
7. MySQL

Orbit



It is an event management mobile application built with flutter. Intended use for this app was, the user can set a goal in this application after registering, also, he can set individual tasks under each goal, separate work list under each task etc. Each task will have a deadline. According to these deadlines, an event will be created on the user's device calendar & the user will get notified accordingly. I was assigned as the team leader for this product. My team had another 2 members, Saidul Islam, Imteaj Sajid. It

was my last project in the intern period. At the end of my internship, this project was on user's review.



Technologies used for this project:

1. Flutter
2. Firebase
3. Laravel
4. Different Packages

3.3 Learning

Things I learned at Luminous Labs:

Mobile Application Development

Most of the time of my internship at Luminous Labs was focused on Mobile Application development. I worked on cross platform mobile application development. Cross platform development is the process of creating mobile applications that are compatible with more than one mobile operating system like Android, IOS. There are some technologies to build mobile applications in such a way. I was instructed to learn flutter for that. Flutter is an open source UI Development Kit developed by Google. It is used to develop cross platform applications for Android, IOS, Linus, Windows, macOS, Web from a single codebase.



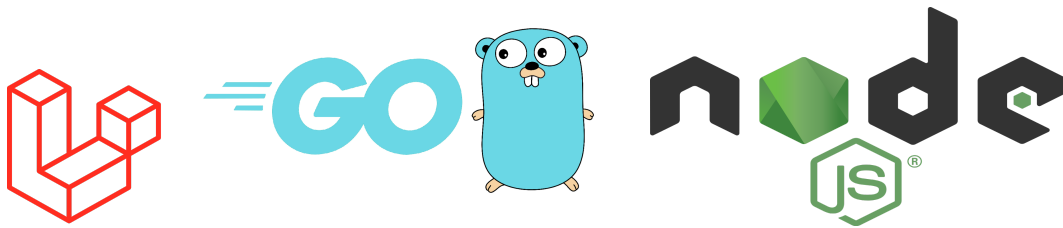
Web Application Development

For web development, I was introduced to the PHP based framework, Laravel. Laravel is a web application framework with expressive, elegant syntax. Laravel attempts to take the pain out of development by easing common tasks used in the majority of web projects, such as authentication, routing, sessions, and caching.

I also got introduced to GoLang & NodeJS. The Go programming language is an open source project to make programmers more productive. Go is expressive, concise, clean, and efficient. Its concurrency mechanisms make it easy to write programs that

get the most out of multicore and networked machines. Go compiles quickly to machine code yet has the convenience of garbage collection and the power of run-time reflection. It's a fast, statically typed, compiled language that feels like a dynamically typed, interpreted language.

Node.js is an open-source and cross-platform JavaScript runtime environment. It is a popular tool for almost any kind of project. Node.js runs the V8 JavaScript engine, the core of Google Chrome, outside of the browser. This allows Node.js to be very performant.



Technologies

1. PHP
2. Dart
3. GoLang
4. Laravel
5. Flutter
6. Javascript
7. NodeJS
8. Firebase
9. Socket.io
10. Google Map
11. Git

Tools

1. Postman
2. VS Code
3. Android Studio
4. XCode
5. Mobile Emulators
6. DartPad
7. Go Playground
8. Bitbucket
9. Github

Chapter 4: Professional Growth

4.1 Teamwork

Though at first half of my internship, I mostly worked solo, I worked with 2 separate teams in the last half. And it was one of the most important experiences of my internship. In any real life work, you have to work with other people. And it is not quite similar to working alone. Different people will have different opinions on a matter, different approach to solve a problem, different time scheduling. But in a team, everything has to go in a synced way. As I didn't have much experience working with more than 2 people, it was quite tough at first to cope up. But as my teammates were quite friendly & kind, they patiently taught me how to manage in these situations.

4.2 Punctuality

Punctuality is very important in professionalism. Luminous Labs takes punctuality very seriously. It is mandatory in Luminous Labs to attend the office just at 10am. Then we had to attend a SCRUM meeting at 10:20am daily. It was very difficult at first to manage this time schedule, but I eventually got used to this.

4.3 Communication

Communication is a very important part to work in a team. Any team with a communication gap will surely suffer. Luminous Labs taught me how to approach senior developers professionally & get help.

4.4 Sharing Knowledge

Every week, Luminous Labs arranges a knowledge sharing meeting. In this meeting, we talk about the new problems we faced this week, how we solved them, and if someone has a better, more optimised solution, they share. It is very important in professional life as it allows us to save time if someone faces the same problem later, to find a more optimised solution, and overall to learn more.

Chapter 5: Conclusion

Internship is a period of learning, putting my academic knowledge to real life work. In my 6 months long internship at Luminous Labs, I learned much about new technologies, tools and professionalism.

I would like to thank Institute of Information & Communication Technology, SUST, for providing me this amazing opportunity to take a glance at the exciting life ahead in the Software Industry. It was one of the most exciting periods of my academic career, where I had the guidance, support & companionship of some very talented people.

Luminous Labs is a rapidly growing software firm of Bangladesh. It was a great opportunity for me to jumpstart my career here. It was probably the busiest period of my life until now. As you can see, I was swamped with tasks, but I enjoyed every moment of it. It was pretty interesting to learn & implement new things continuously. People at Luminous Labs taught me so much in this period, I can't thank them enough. And I think they are pretty happy with my performance too, as they have already offered me full-time employment in their company.

From my experience, I would recommend Luminous Labs as a future internship partner of IICT, SUST. Luminous Labs provides a great opportunity to learn & work in real projects, which makes Luminous Labs different from other software companies. If a developer is capable, Luminous Labs doesn't delay to give him the opportunity to utilise his skill in real work. One can start working on a big project very quickly if he has the necessary skills, which makes growth pretty fast for junior developers & freshers. This is something that impressed me a lot in Luminous Labs.

In the end, I can finish by saying that this internship was a great experience for me. This opportunity has given me the chance to use my knowledge solving real life problems, made me more confident, time efficient & professional. I hope this experience will help me greatly in my future career as a software engineer.